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Rajiv Gandhi University of Knowledge Technologies- RK Valley- AP

Academic Year 2023– 24: Semester 2 - Mid Semester Test-2

Civil - Structural Analysis

R20 Batch &Regular Mode of Examinations

R210519

Year:ENGG-2 **Date of Examination: 06-04-2024**

Time:60 Minutes **Max Marks:15**

3X5 Marks= 15

The Question Paper Consists of 6 questions in 3 sets. Answer 1 question from each set
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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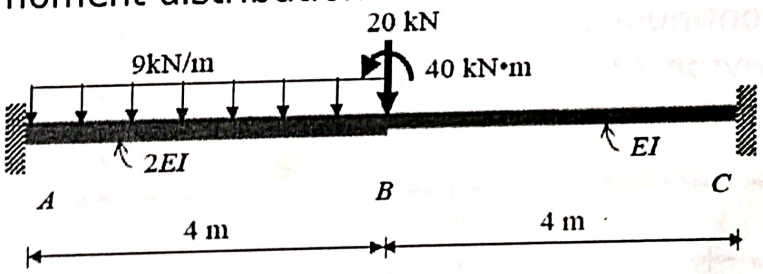
1. State distribution theorem with neat diagram

(OR)

2. What will be the additional shear and moments induced in a fixed beam, when one of its supports settle by δ.

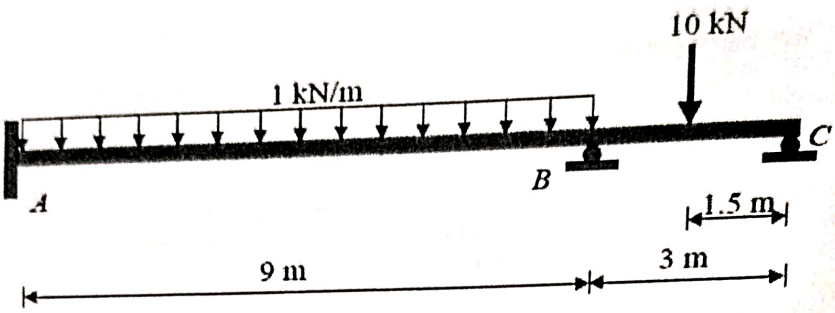
3. Analyze the structure shown in the figure below using moment distribution method



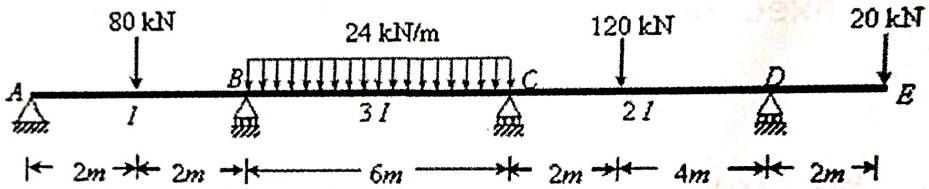
(OR)

4. Analyze the structure shown in the figure below using Clapeyron's theorem of three moments.

Note: Flexural rigidity (EI) same for all the members

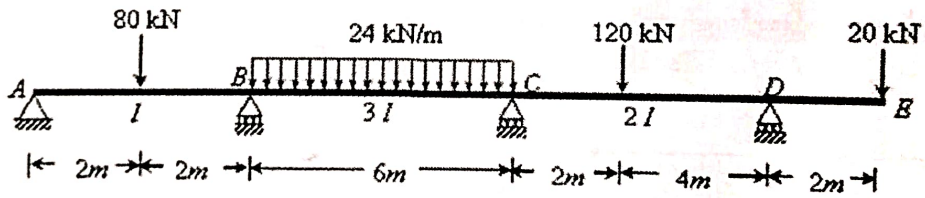


5. Analyze the continuous beam shown in the figure below using moment distribution method.



(OR)

6. Analyze the continuous beam shown in the figure below using Clapeyron's theorem of three moments.



Rajiv Gandhi University of Knowledge Technologies- RK Valley- AP

R210517

Academic Year 2023-24: Semester II - Mid Semester Test-2

Civil Engineering - 20CE2206- Water Resources Engineering

R20 Batch& Regular Mode of Examinations

Date of Examination: April, 2024 — 4 , PM

Year:ENGG-2

Time:~~45~~ Minutes 60

Max Marks: 15

3X5 Marks= 15

The Question Paper Consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions
Note: Usage of any data/ Scientific calculator/code book may be permitted/not permitted (if any)

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1. A storm with 10 cm of precipitation produced a direct runoff of 5.8 cm. The duration of the rainfall was 16 hours and its time distribution is given below. Estimate the ϕ -index of the storm.

Time from start (h)	0	2	4	6	8	10	12	14	16
Cumulative Rainfall (cm)	0	0.4	1.3	2.8	5.1	6.9	8.5	9.5	10.0

OR

2. The mass curve of rainfall of duration 100 min is given below. If the catchment had an initial loss of 0.6 cm and ϕ –Index of 0.6 cm/hr, calculate the total surface runoff from the catchment.

Time from Start (min)	0	20	40	60	80	100
Cumulative Rainfall (cm)	0	0.5	1.2	2.6	3.3	3.5

3. With the help of typical annual hydrograph describe the salient features of (i) Perennial Streams ii) Intermittent Streams and iii) Ephemeral Streams

OR

4. Explain in brief about flow duration curve? Define Drought? What is the different Classification of Drought?

5. Explain in brief about elements of flood hydrograph? Define a unit hydrograph. Explain two basic assumptions made in the derivation of unit hydrograph.

OR

6. Following are the ordinates of a 4-hr unit hydrograph. Using this, derive the ordinates of a 12-hr unit hydrograph

Time (hrs)	0	4	8	12	16	20	24	28	32
4 hr UH (m^3/s)	0	20	80	130	90	25	15	5	0

Rajiv Gandhi University of Knowledge Technologies- RK Valley- AP

Academic Year 2023-24: Semester II - Mid Semester Test-2
Civil Engineering - 20CE 2203 – Soil Mechanics
R20 Batch& Regular Mode of Examinations

R210517

Year:ENGG-2 | **Date of Examination: April, 2024** – 3
Time:45 Minutes | **Max Marks: 15**

3X5 Marks= 15

The Question Paper Consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions
Note: Usage of any data/ Scientific calculator/code book may be permitted (if any)
Assume suitable data if missing

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1. a) Falling head permeability test was conducted on a soil sample of 4 cm dia & 18 cm length. The head fell from 1.0 m to 0.4 m in 20 minutes. If the cross section area of stand pipe is 1 sq.cm, determine k. (3 M)

b) k of the soil at e of 0.7 is 4 x 10^(-4) cm/s. Estimate its value at e of 0.5 (2 M)

OR

2. a) Calculate the height of rise of capillary water having flowing passage of 0.0002 m. (2 M)

b) Define quick sand condition and derive the equation for quick sand condition. (3 M)

3. A sand profile consists of a surface layer of sand 3.5 m thick (1.65 g/cc), an intermediate layer of clay of 3 m thick (1.95 g/cc) and the bottom layer of gravel 3.5 m thick (1.925 g/cc). Water table is at the upper surface of clay layer. Determine the effective pressure at various levels immediately after placement of surcharge load of 58.86 kN/m^2 at the ground surface?

OR

4. Explain the factors affecting coefficient of permeability of soils.

5. Define flow line and equipotential line. Explain any 3 characteristics of flow net.

OR

6. Determine vertical stress at a point P which is 3m below and at a radial distance of 3 m from a vertical load of 100 kN. Use Westergaard's theory.

Rajiv Gandhi University of Knowledge Technologies- RK Valley- AP

Academic Year 2023-24: Semester II - Mid Semester Test-2

Civil Engineering - 20CE2201- Hydraulic Engineering R210517

R20 Batch& Regular Mode of Examinations

Year:ENGG-2 **Date of Examination: April, 2024** — 2

Time:45 Minutes **Max Marks: 15**

3X5 Marks= 15

The Question Paper Consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

Note: Usage of any data/ Scientific calculator/code book may be permitted/not permitted (if any)

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1. Derive the Chezy Equation for Uniform Flow?

OR

2. Derive the differential equation of Gradually Varied Flow?

3. A Trapezoidal channel is 10 m wide and has a slope of 1.5H : 1V. The bed slope is 0.0003. The channel is lined with smooth concrete of n = 0.012.

a) Compute the mean velocity and discharge for a depth of flow of 3.0 m.

b) Find the bottom slope necessary to carry a discharge of 50 m^3/s at a depth of 3.0 m

OR

4. A slightly rough lined trapezoidal channel with manning's roughness coefficient of n= 0.017 and carrying a discharge of 25 m^3/s is to have a longitudinal slope of 0.0004. Determine the proportions of an efficient trapezoidal channel section having a side slope of 1.5H : 1V.

5. Show that for a trapezoidal channel of given area of flow, the condition of maximum flow requires that hydraulic mean depth is equal to one half the depth of uniform flow.

OR

6. Using the basic differential equation of GVF, show that dy/dx is positive for S1, M3 and S3 profiles